

MARKED SYMMETRIC S -ADIC SHIFTS: EXACT MEASURE INTERVALS AND A SUMMABILITY TRANSITION

ANONYMOUS

ABSTRACT. Let $(a_n)_{n \geq 1}$ be positive integers and consider the constant-length directive morphisms

$$\sigma_n(0) = 0^{a_n+1}1, \quad \sigma_n(1) = 01^{a_n+1}.$$

We give a word-level analysis of the resulting binary S -adic shift. The word 10 occurs exactly across image boundaries, so it supplies an explicit marker at every level. This yields unique desubstitution, two clopen towers at each scale, minimality, and aperiodicity without invoking an abstract recognizability theorem. If

$$R = \prod_{n \geq 1} \frac{a_n}{a_n + 2},$$

then the invariant-measure simplex is affinely homeomorphic to $[-R, R]$ by the symbol bias $b(\mu) = \mu([0]) - \mu([1])$. Thus the actual frequency of 0 ranges over an interval of radius $R/2$, not R . The system is uniquely ergodic exactly when $\sum_n 1/a_n$ diverges; otherwise it has exactly two ergodic measures. For $a_n = n$, the finite-prefix bias radius is $R_N = 2/((N+1)(N+2))$ and tends to zero. For $a_n = n^2$, the limiting bias radius is $\pi\sqrt{2}/\sinh(\pi\sqrt{2})$. Literal marker enumeration and independent exact incidence arithmetic verify the registered finite controls.

1. INTRODUCTION

A nonstationary substitution can be primitive at every level and still fail to be uniquely ergodic. The obstruction is not visible in any one incidence matrix: it is the accumulated contraction, or lack of contraction, along the entire directive sequence. The binary family in this note isolates that mechanism in a form where both the symbolic cutting and the measure simplex are explicit.

Fix positive integers a_n and set

$$(1) \quad \sigma_n(0) = 0^{a_n+1}1, \quad \sigma_n(1) = 01^{a_n+1}.$$

Both images have length $a_n + 2$, begin with 0, and end with 1. More decisively, neither image contains 10, while every concatenation boundary does. The same two-letter word therefore marks every level- n cut. This first anomaly turns a matrix construction into a concrete symbolic system.

The contribution of the note is an exact, self-contained package for this marked realization.

- (1) We prove all-level recognizability directly from the marker 10 and use the resulting nested cuts to establish minimality and aperiodicity.
- (2) We identify invariant probabilities with the inverse limit of the two normalized incidence simplices. The proof constructs every measure from compatible two-tower weights and does not assume unique ergodicity.
- (3) We diagonalize that inverse limit in the physical symbol-bias coordinate. Its radius is the infinite product R , whereas the radius of the 0-frequency interval is $R/2$. This also gives the two ergodic endpoints and the closed constants for $a_n = n$ and $a_n = n^2$.

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Prior-results boundary. The S -adic formalism and incidence-cone approach to frequencies are standard; see the survey of Berthé and Delecroix [4]. Berthé, Steiner, Thuswaldner, and Yassawi develop recognizability for sequences of morphisms [5], while Bédaride, Hilion, and Lustig develop the tower description of invariant measures [2] and the associated measure-transfer framework [3].

Most importantly, Ferenczi, Fisher, and Talet already treated the unique-ergodicity criterion for adic systems with symmetric matrix sequence $(\begin{smallmatrix} a_n & 1 \\ 1 & a_n \end{smallmatrix})$ [6]. Arbulú, Durand, and Espinoza explicitly record the same summability criterion in their constant-length S -adic discussion [1]. Accordingly, the equivalence with $\sum 1/a_n = \infty$ below is a specialized rederivation, not a priority claim. The latter paper also contains close two-letter constant-length examples in which a reciprocal-sum condition separates unique ergodicity from a two-measure regime. Thus neither the inverse-limit mechanism nor the qualitative phase-transition pattern is claimed as new. What is recorded here is the full word-level realization (1): its elementary marker proof, its complete affine measure interval in the symbol-frequency coordinate, and two exact product specializations. A bounded search through 28 August 2026 did not locate this precise combined package. That search is not a novelty proof, and external release remains on hold.

2. DIRECTIVE LANGUAGE AND THE MARKER

Let $\mathcal{A} = \{0, 1\}$. For $1 \leq m < n$, write

$$\tau_{m,n} = \sigma_m \sigma_{m+1} \cdots \sigma_{n-1}, \quad h_{m,n} = \prod_{j=m}^{n-1} (a_j + 2),$$

and put $\tau_{n,n} = \text{id}$ and $h_{n,n} = 1$. Composition is read from right to left, so $\tau_{m,n}$ maps level- n letters to level- m words.

Definition 2.1. For $m \geq 1$, the level- m tail shift $X^{(m)} \subseteq \mathcal{A}^{\mathbb{Z}}$ consists of the sequences all of whose finite factors occur in $\tau_{m,n}(c)$ for some $n > m$ and $c \in \mathcal{A}$. The shift studied in this note is $X = X^{(1)}$, with the left shift T .

Every incidence matrix is strictly positive, so a word that occurs at one level occurs inside both sufficiently high supertile types. The directive language is therefore recurrent and extendable, and the compact subshifts in definition 2.1 are nonempty.

The next proposition is the symbolic hinge of the paper.

Proposition 2.2 (the 10 marker). *Put $L_m = a_m + 2$. For every $y \in X^{(m+1)}$, the occurrences of 10 in $\sigma_m(y)$ start exactly at the positions*

$$L_m \mathbb{Z} - 1.$$

Consequently every $x \in X^{(m)}$ has a unique centered representation

$$(2) \quad x = T^r \sigma_m(y), \quad 0 \leq r < L_m, \quad y \in X^{(m+1)}.$$

The cut phase r is read from the residue class of the occurrences of 10, and each block between consecutive cuts uniquely determines its preimage letter.

Proof. Set $a = a_m$. The two image words are

$$A = 0^{a+1}1, \quad B = 01^{a+1}.$$

Neither contains 10. Both start with 0 and end with 1, so every image boundary contributes exactly one occurrence of 10. In an unshifted image $\sigma_m(y)$, those boundaries lie between coordinates $kL_m - 1$ and kL_m . This proves the first assertion.

For existence in (2), approximate a point of $X^{(m)}$ by shifted factors of higher supertiles. Pass to a subsequence on which the natural level- m cut has one fixed residue modulo L_m . Compactness of $\mathcal{A}^{\mathbb{Z}}$ gives a limiting preimage y , and its finite factors belong to the level- $(m+1)$ directive language. Hence $y \in X^{(m+1)}$.

In any representation, the starts of 10 form one residue class modulo L_m . That residue determines r uniquely. Once the cuts are known, every intervening length- L_m block is either A or B , which are distinct and decode to 0 and 1, respectively. Thus y is unique as well. $\square$

Iterating [proposition 2.2](#) gives a unique level- n cut phase modulo

$$(3) \quad h_n := h_{1,n} = \prod_{j=1}^{n-1} (a_j + 2).$$

This is stronger than an almost-everywhere cutting statement: every point has one cut phase at every level.

3. MINIMALITY AND APERIODICITY

The marker supplies the cuts; strict positivity supplies recurrence.

Theorem 3.1 (minimal aperiodic realization). *For every sequence of positive integers (a_n) , the subshift (X, T) is minimal and has no periodic point.*

Proof. Let $u \in \mathcal{L}(X)$. Choose n and $c \in \mathcal{A}$ such that u occurs in the level- n supertile $\tau_{1,n}(c)$. Each word $\sigma_n(b)$ contains both letters for $b \in \mathcal{A}$. Therefore each level- $(n+1)$ supertile $\tau_{1,n+1}(b)$ contains $\tau_{1,n}(c)$ and hence contains u . By [proposition 2.2](#), every $x \in X$ is tiled by level- $(n+1)$ supertiles of common length h_{n+1} . Consecutive occurrences of u in x are thus separated by at most $2h_{n+1}$. Every allowed word is uniformly recurrent, which is equivalent to minimality for a subshift.

Now suppose $T^p x = x$ for some $p > 0$. The unique level- n cut set of x is one residue class modulo h_n . Shifting by p translates that cut set, while periodicity and uniqueness require the translated set to equal the original one. Hence h_n divides p for every n . Since $h_n \geq 3^{n-1}$, no positive integer has this property. Thus X is aperiodic. $\square$

The proof also explains why recognizability cannot be discarded as a technical convenience: the divisibility argument uses the uniqueness, not merely the existence, of every supertile phase.

4. TWO TOWERS AND INVARIANT MEASURES

For $n \geq 1$ and $c \in \mathcal{A}$, let

$$W_{n,c} = \tau_{1,n}(c), \quad B_{n,c} = \tau_{1,n}([c]_{X^{(n)}}).$$

The set $B_{n,c}$ consists of the points whose level- n cut is at the origin and whose supertile there has type c . The marker makes these sets clopen. The families

$$(4) \quad \mathcal{P}_n = \{T^j B_{n,c} : c \in \mathcal{A}, 0 \leq j < h_n\}$$

are two-tower clopen partitions, and $\mathcal{P}_{n+1}$ refines $\mathcal{P}_n$.

With rows indexed by output letters and columns by input letters, the incidence and normalized incidence matrices are

$$(5) \quad M_n = \begin{pmatrix} a_n + 1 & 1 \\ 1 & a_n + 1 \end{pmatrix}, \quad P_n = \frac{M_n}{a_n + 2}.$$

Write $\Delta = \{(p_0, p_1)^\top : p_c \geq 0, p_0 + p_1 = 1\}$.

Theorem 4.1 (two-tower affine bijection). *For a T -invariant probability μ , define*

$$(6) \quad p_c^{(n)} = h_n \mu(B_{n,c}), \quad c \in \mathcal{A}.$$

Then $p^{(n)} \in \Delta$ and

$$(7) \quad p^{(n)} = P_n p^{(n+1)}.$$

The map

$$(8) \quad \mu \longmapsto (p^{(n)})_{n \geq 1}$$

is an affine bijection from $\mathcal{M}(X, T)$ onto

$$\varprojlim (\Delta, P_n) = \{(p^{(n)} : p^{(n)} \in \Delta, p^{(n)} = P_n p^{(n+1)})\}.$$

Proof. The h_n floors above a fixed base have equal measure. Since (4) partitions X , the two numbers in (6) sum to one. A level- $(n+1)$ supertile of type b contains $M_n(c, b)$ level- n supertiles of type c . Refining the bases, using shift invariance, and recalling $h_{n+1} = (a_n + 2)h_n$ gives

$$p_c^{(n)} = \sum_{b \in \mathcal{A}} \frac{M_n(c, b)}{a_n + 2} p_b^{(n+1)},$$

which is (7).

We prove that no information is lost and that every compatible sequence is realized. For a finite word u , let $N_u(w)$ count occurrences of u whose full support lies inside the finite word w . Given compatible tower weights, set

$$(9) \quad \Gamma_n(u) = \sum_{c \in \mathcal{A}} p_c^{(n)} \frac{N_u(W_{n,c})}{h_n}.$$

The level- $(n+1)$ words are concatenations of level- n words. By (7), occurrences wholly contained in one constituent block contribute exactly $\Gamma_n(u)$. Only occurrences crossing a constituent boundary remain. There are at most $(a_n + 1)(|u| - 1)$ of those in each level- $(n+1)$ supertile, and hence

$$(10) \quad |\Gamma_{n+1}(u) - \Gamma_n(u)| \leq \frac{|u| - 1}{h_n}.$$

Because $h_n \geq 3^{n-1}$, the sequence $\Gamma_n(u)$ converges.

Define $\mu([u]) = \lim_n \Gamma_n(u)$. Nonnegativity is immediate. Summing over all one-letter extensions fails at finite level only for occurrences touching an endpoint of $W_{n,c}$; the normalized error is at most $1/h_n$. Taking limits gives both cylinder consistency identities and shift invariance. Thus the values extend to a T -invariant Borel probability. Words outside the directive language have zero value, so the measure is supported on X .

Conversely, decompose a cylinder $[u]$ over the towers (4). The floors on which u stays inside one supertile give (9); the total measure of the boundary floors is at most $(|u| - 1)/h_n$. Therefore every invariant μ satisfies $\mu([u]) = \lim_n \Gamma_n(u)$ and is determined by its tower weights. For the measure constructed from compatible weights, counting the marked level- n bases inside higher supertiles recovers $h_n \mu(B_{n,c}) = p_c^{(n)}$; high-super-tile endpoints have vanishing density. The two constructions are inverse and affine. $\square$

General tower and measure-transfer results cover much wider classes [2, 3]. The proof above is included to make the normalization and the direction of (7) auditable for this family.

5. BIAS INTERVAL AND THE PHASE TRANSITION

The matrices in (5) share the eigenvectors $(1, 1)$ and $(1, -1)$. Put

$$(11) \quad \rho_n = \frac{a_n}{a_n + 2}, \quad R_N = \prod_{n=1}^N \rho_n, \quad R = \prod_{n \geq 1} \rho_n.$$

For $t \in [-1, 1]$, write

$$v(t) = \frac{1}{2} \begin{pmatrix} 1+t \\ 1-t \end{pmatrix}.$$

Direct multiplication gives the key one-line identity

$$(12) \quad P_n v(t) = v(\rho_n t).$$

Theorem 5.1 (exact invariant-measure interval). *The map*

$$(13) \quad \Theta : \mathcal{M}(X, T) \longrightarrow \mathbb{R}, \quad \Theta(\mu) = b(\mu) := \mu([0]) - \mu([1])$$

is an affine homeomorphism from $\mathcal{M}(X, T)$ onto $[-R, R]$. Consequently

$$(14) \quad \{\mu([0]) : \mu \in \mathcal{M}(X, T)\} = \left[\frac{1-R}{2}, \frac{1+R}{2} \right].$$

Thus R is the bias radius, while $R/2$ is the radius of the symbol-frequency interval around $1/2$.

Proof. Let $b_n = p_0^{(n)} - p_1^{(n)}$ be the level- n tower bias. By (12), compatibility is equivalent to

$$(15) \quad b_n = \rho_n b_{n+1}.$$

In particular, $b_1 = R_N b_{N+1}$, so $|b_1| \leq R_N$ for every N and $|b_1| \leq R$.

Suppose first that $R > 0$. For any $t \in [-R, R]$, set

$$b_1 = t, \quad b_n = \frac{t}{R_{n-1}} \quad (n \geq 2).$$

Since $R_{n-1} \geq R$, all b_n lie in $[-1, 1]$, and (15) holds. This produces one compatible sequence, and (15) shows that it is the only one with first bias t .

If $R = 0$, then for each fixed n the tail product $\prod_{j=n}^N \rho_j$ tends to zero. Iterating (15) gives

$$|b_n| \leq \prod_{j=n}^N \rho_j,$$

so every b_n is zero. The inverse limit is then a singleton. Theorem 4.1 proves bijectivity in both cases. The map is continuous, and a continuous bijection from the compact measure simplex to a Hausdorff interval is a homeomorphism. Finally, $\mu([0]) = (1 + b(\mu))/2$, which proves (14). $\square$

Corollary 5.2 (summability transition). *The shift (X, T) is uniquely ergodic if and only if*

$$(16) \quad \sum_{n \geq 1} \frac{1}{a_n} = \infty.$$

If the series converges, then $R > 0$ and (X, T) has exactly two ergodic probability measures, corresponding to the biases $-R$ and R .

Proof. Write $\rho_n = 1 - y_n$ with $y_n = 2/(a_n + 2)$. Since $0 < y_n \leq 2/3$,

$$y_n \leq -\log(1 - y_n) \leq \frac{y_n}{1 - y_n} \leq 3y_n.$$

Hence $R > 0$ exactly when $\sum_n 1/(a_n + 2) < \infty$. For $a_n \geq 1$,

$$\frac{1}{3a_n} \leq \frac{1}{a_n + 2} \leq \frac{1}{a_n},$$

so this is equivalent to convergence of $\sum_n 1/a_n$.

By theorem 5.1, $R = 0$ gives one invariant probability. When $R > 0$, the invariant-measure simplex is a nondegenerate interval, whose only extreme points are its two endpoints. Ergodic invariant probabilities are exactly the extreme points, proving the last assertion. $\square$

As noted in the introduction, the yes/no criterion in corollary 5.2 is already present at the adic matrix level in [6]; the contribution of theorem 5.1 is the explicit affine interval for the marked symbolic realization and the conversion to physical one-symbol frequencies.

6. TWO CLOSED DIRECTIVE SEQUENCES

The general product becomes elementary for two natural choices of a_n .

Corollary 6.1 (linear directive). *If $a_n = n$, then*

$$(17) \quad R_N = \prod_{n=1}^N \frac{n}{n+2} = \frac{2}{(N+1)(N+2)}.$$

Thus $R = 0$, the system is uniquely ergodic, and its unique measure has $\mu([0]) = \mu([1]) = 1/2$.

Proof. Cancel $1 \cdot 2 \cdots N$ against $3 \cdot 4 \cdots (N+2)$ in the finite product. The remaining assertions follow from [theorem 5.1](#). $\square$

Corollary 6.2 (quadratic directive). *If $a_n = n^2$, then*

$$(18) \quad R = \prod_{n \geq 1} \frac{n^2}{n^2 + 2} = \frac{\pi\sqrt{2}}{\sinh(\pi\sqrt{2})} = 0.104528666176957\dots$$

There are exactly two ergodic probabilities. Their 0-symbol frequencies are

$$(19) \quad \frac{1}{2} \left(1 \pm \frac{\pi\sqrt{2}}{\sinh(\pi\sqrt{2})} \right) = 0.4477356669\dots, \quad 0.5522643331\dots$$

Proof. Euler's product

$$\frac{\sinh(\pi x)}{\pi x} = \prod_{n \geq 1} \left(1 + \frac{x^2}{n^2} \right)$$

at $x = \sqrt{2}$ gives (18). The measure statements and the factor $1/2$ in (19) follow from [theorem 5.1](#) and [corollary 5.2](#). $\square$

The contrast is sharp at the first level: the linear directive loses every nonzero bias in the infinite tower limit, whereas the quadratic directive retains a bias interval of positive width. No finite prefix distinguishes the two regimes categorically; the infinite product does.

7. EXACT CONTROLS AND SCOPE

The registered script has two independent exact lanes. The literal lane enumerates every binary preimage word of lengths one through seven for $1 \leq a \leq 9$. It checks both linearly and cyclically that 10 occurs exactly at image boundaries, that consecutive cyclic markers have gap $a + 2$, and that the cut blocks decode to the original letters. The matrix lane uses `fractions.Fraction` to verify (12) for individual matrices, all directives of lengths one through six over $a \in \{1, 2, 3, 4\}$, and compatible finite inverse-limit biases. It also verifies (17) exactly through $N = 200$ and the quadratic partial-product recurrence through $N = 100$.

These controls guard marker endpoints, matrix orientation, normalization, and the factor-of-two conversion from bias to frequency. They do not replace the compactness argument in [proposition 2.2](#), the all-level measure construction in [theorem 4.1](#), or the infinite-product proof.

This manuscript is an internal theorem record. It does not claim that the summability criterion is new, and it does not make a global claim about all constant-length S -adic systems. Public posting, submission, and priority language remain on hold pending a specialist comparison with the adic and Toeplitz literature.

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